

Carolina Del Pilar Díaz

📍 Córdoba, Argentina ✉ caropilardiaz@gmail.com ☎ 0351 15-542-8714 🌐 carolinadelpilardiaz

Summary

AI/ML Engineer, Senior Data Scientist, and Applied Statistician (Economist) with 20+ years of experience operating as a Strategic AI/ML Consultant & Forward Deployed Engineer (FDE) for global enterprises. I specialize in architecting end-to-end AI products and solutions using Python, Machine Learning, and Deep Learning. I build mathematically robust Multi-Agent Systems (GenAI, RAG, MCP) that prevent model hallucinations in regulated sectors. I have architected and shipped full-stack AI applications, combining the power of Python with the rigor of Advanced Statistics, Causal Inference, and MLOps on on-premises, hybrid, and cloud infrastructures across Fintech, AgTech, and Healthcare.

Experience

IT Patagonia (Consultant FDE), Senior AI Engineer

Buenos Aires, Argentina

June 2024 – present

- Forward Deployed Engineer (FDE) & Technical Lead: Directed the architecture and deployment of Generative AI, RAG, and Multi-Agent solutions for enterprise clients (Banco Galicia, ICBC), defining technical roadmaps and optimizing cloud infrastructure.
- Agentic AI & Complex Workflows: Architected a 33-node LangGraph state-machine for automated legacy codebase documentation (Patagon.IA) and engineered a 100% asynchronous LlamaIndex RAG pipeline with parallel retrieval for a real-time sales assistant (Facturante).
- LLM Cost Optimization & Resiliency: Implemented state-checkpointing to avoid API reprocessing in long-running LLM pipelines, enforced strict token budgets (ThinkingConfig), and designed dynamic routing between cloud (AWS Bedrock) and local (Ollama) foundational models.
- Quality Assurance & Serverless Deployments: Built an advanced 'LLM-as-a-judge' multi-evaluator system using majority voting to eliminate hallucinations. Shipped containerized (Docker) hybrid deployments and serverless MLflow PyFunc inference models on Databricks.

ITP Patagonia - ICBC, Senior Data Scientist

Buenos Aires, Argentina

June 2024 – July 2025

- Developed predictive models for payroll clients and mutual fund investment using AWS SageMaker, AutoML, Canvas, and Neural Networks/XGBoost. Impact: enabled efficient predictions and insights for improved decision-making.
- Built an anomaly detection model using Neural Networks (AENN), Isolation Forest, and SHAP for interpretability using Azure. Impact: integrated into fraud monitoring systems for efficient detection and prevention.
- Currently developing a fraud detection model with AWS Fraud Detector, leveraging Decision Trees, Neural Networks, Time Series Analysis, and Custom Fraud Prediction. Impact: significantly reduced client account lockouts (Project Owner).

Nubiral S.A., Senior Data Scientist. COE, Commission of Expert. Generative Models.

Buenos Aires, Argentina

Oct 2023 – May 2024

- Airline solution: Integrated AWS Rekognition, Lambda, S3, LangChain, and OpenAI GPT for Computer Vision and Generative AI, improving object detection accuracy and increasing precision at Aerolíneas Argentinas.
- Developed a Post-Call Analytics solution using AWS services (Transcribe, Comprehend, Lambda, QuickSight) to analyze customer interactions, extract insights, and generate actionable reports to improve customer service at Telecom S.A. Impact: Improved efficiency.
- Implemented Azure Cognitive Search and Form Recognizer for efficient PDF document management at Banco Macro, enhancing data extraction and document accessibility across the organization. Impact: Streamlined operations.

Pi Data Strategy & Consulting , Semi-Senior Data Scientist, Statistician, Economist		Córdoba, Argentina
- Consulting and advice: Created data regression models, predictive data modelling, and analyzed data mining algorithms, creating end-to-end pipelines to deploy on Azure. - Applied Machine Learning, Speech Tech (Sequence Matcher), and NLP using IBM technology. Performed EDA, statistical analysis, and built neural networks for Pernod Ricard, Don Mario (GDM Seeds), and Anheuser-Busch InBev. - Developed sales forecasting models for georeferenced datasets and multinomial predictive models for customer satisfaction levels (Project Owner).		Mar 2021 – Oct 2023
National Institute of Agriculture Technology (INTA) , Lead Statistician / Head of Statistical Research		Manfredi, Córdoba, Argentina
- Designed and implemented experimental designs and biometric models for all research projects in ecophysiology, plant breeding, and environmental sustainability, serving as Head of Statistical Research. - Applied neural networks, deep learning, and computer vision to UAV and agricultural machinery, enhancing crop health monitoring, soil analysis, pest detection, and precision farming.		Jan 2012 – Dec 2021

Education

Postgraduate Master's Degree	National University of Córdoba , Applied Statistics	Córdoba, Argentina Jan 2006 – Dec 2008
Diploma	National University of Córdoba , Data Science Machine Learning and applications	Córdoba, Argentina Jan 2019 – Dec 2020
Bachelor's degree	National University of Córdoba , Economics	Córdoba, Argentina Jan 1993 – Dec 1998

Skills

Agentic & Generative AI: Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, AutoGen, Semantic Kernel, RAG, MCP, Prompt Engineering, LLMs (Gemini, GPT, Claude, Ollama, Hugging Face)

Machine Learning & Stats: Deep Learning, Computer Vision, Causal Inference, Bayesian Statistics, Econometrics, Anomaly Detection, Time Series, MLOps

Cloud & Big Data: AWS (SageMaker, Lambda, Rekognition, Fraud Detector), Azure AI Services, Snowflake, Databricks, Apache Spark, Python, R, SAS

Languages: Spanish (Native), English (Intermediate), Portuguese (Basic)

Projects

PATAGON.IA — Automated COBOL Legacy Documentation	Jan 2026
33-node LangGraph pipeline on AWS Bedrock generating BRD/TRD docs from COBOL source. Led prompt engineering of ~23 sections (BRD/TRD), achieving 20/20 sections verified with 0 pipeline errors. Implemented a 3-judge LLM panel with majority voting and anti-hallucination filters against a deterministic golden-truth (judge integrated as node n34). Managed Human-in-the-Loop flow (Excel + Google Drive + knowledge packs) and multi-model adaptation (DeepSeek R1 / Qwen / GLM) with reproducible runs (temperature=0). IT Patagonia.	

Banco Galicia — AI Architecture & Observability Assessment

Jan 2026

Observability and regulatory compliance strategy designed during an architectural audit of two production AI systems (multi-agent orchestrator and RAG agent). Diagnosed gaps such as missing P50/P95/P99 latency metrics, log contexts, health checks, and tracing. Proposed a target architecture integrating Prometheus (HTTP/Kafka metrics), Grafana, Azure Application Insights, and OpenTelemetry + Jaeger for distributed tracing. Designed hallucination auditing using RAGAS metrics (faithfulness, answer relevancy) and alerting for SLAs. Aligned system requirements to BCRA (Com. 'A' 8280) and CNV regulatory traceability frameworks. IT Patagonia.

Banco Macro — Nightly Batch Completion Predictor (OPTI+)

2025-2026

Predictive model for nightly batch execution end-time at Banco Macro, replacing static heuristics with statistical learning and communicating predictions as time windows with natural-language explanations. Developed and evaluated machine learning models including Linear Regression, Decision Trees, Random Forest, HistGradientBoosting, and Prophet time-series over 485 real records (optimized MAE to ~39 min). Designed an instance-based estimator (k-NN weighted by day type, volume, and day of week) with robust median estimation, standard deviation outlier filtering, and asymmetric prediction intervals based on distribution skewness. Applied asymmetric loss functions aligned to a 15-minute SLA, achieving 99.0% SLA compliance in a 105-day leave-one-out backtest. Shipped as a deployable product with FastAPI, React + Vite, Node/Express proxy, and Docker, integrated in real-time with the OPTI API and Gemini-powered explanations. IT Patagonia.

ICBC — Banking Customer Segmentation (RFM)

Jan 2025

RFM (Recency, Frequency, Monetary) customer segmentation based on a customer-360 view with feature engineering on ~250 features (products, balances, card usage, risk scores). Engineered data preparation workflows in AWS SageMaker Canvas and Data Wrangler (using PySpark) over Amazon Athena tables, performing monthly filtering and exporting to S3 in Parquet format. Developed feature pipelines including robust-sigma outlier handling, median/categorical imputation, and ordinal/one-hot/similarity encoding (MinHash LSH + n-grams). Applied STL/LOESS time-series decomposition (statsmodels) over transaction history (~1 million records monthly per client). IT Patagonia.

Facturante — WhatsApp Sales AI Agent

Jan 2026

Conversational sales agent deployed as an MLflow PyFunc model on Databricks Model Serving with RAG over product manuals (Databricks Vector Search) and Gemini. Architected on Databricks (Model Serving, Vector Search, Delta Lake) with webhook handoff and multi-phase sales conversation flow. Designed a testbed with automated evaluation (LLM-as-a-judge over a golden dataset), consistency scoring across runs, and a conversation viewer in Databricks Lakeview for auditing. IT Patagonia.

ITP Flokzu — Agentic Document Platform

Jan 2026

Two-agent platform: structured extraction of invoices/receipts from PDF with amount validation, and eligibility evaluation for the Knowledge Economy Regime. Deployed on AWS Lambda (Python 3.13) + AWS Bedrock (Claude 3.5 Sonnet / Haiku) using Strands framework with versioned prompts in S3. Extracted PDF content (PyMuPDF, pdfplumber, pypdf) to structured JSON outputs, integrated via HTTP events with Flokzu BPM, and configured logging and monitoring via Amazon CloudWatch. IT Patagonia.

ICBC — Legacy Modernization VB6 → C#/.NET 8 Modernization of the judicial orders banking system (AFIP integration and BCRA import) migrating from Visual Basic 6 (VB6) to a modern layered architecture. Designed a layered system (Core / Data / Business / WinForms / AfipConnection) using .NET 8, Entity Framework Core, Dependency Injection, async/await, Serilog, and IBM DB2 (IBM.Data.DB2.Core) database connectivity. IT Patagonia.	Jan 2026
Interbanking — Conversational Operations Agent (POC) Technical proposal for a governed SQL + RAG agent on Databricks + S3 (Lakehouse Medallion, Unity Catalog RLS/CLS, Model Serving, Vector Search, grounded RAG). IT Patagonia.	Jan 2026
Universal Assistance — QA Test Automation Web application generating Playwright (TypeScript) test scripts from natural-language requirements using conversational AI, featuring a results dashboard. Developed with React 19, TypeScript, and Vite, leveraging Google Gemini 2.5 Flash for automated code generation oriented to @playwright/test. IT Patagonia.	Jan 2026
Banco Macro — Intelligent Assets Platform Web application that loads asset data (CSV/Excel), analyzes it with conversational AI, and displays real-time reports and visualizations, including cost projections. Built with React 19, TypeScript, and Vite. Integrated Google Gemini 2.5 Flash for automatic column mapping and contextual chat, and used Recharts and math.js for reporting and 3-month cost forecasting. IT Patagonia.	Jan 2026
CPCE Legalizations Web Assistant Web application assisting the professional legalizations process: conversational chat on regulatory standards (RT 37, Res. 34/2024, checklists) and PDF/DOCX document upload with automated analysis of errors, omissions, and inconsistencies. Developed with React 19, TypeScript, Vite, and Tailwind, using mammoth (DOCX) and pdfjs-dist (PDF) for document extraction, and Gemini 2.5 Flash with an embedded regulatory knowledge base.	Jan 2026
Car One — Vehicle Price Intelligence PoC Multi-agent Python Proof of Concept (PoC) that collects vehicle listings from multiple sources, validates the data, and outputs a pricing recommendation for a target vehicle. Engineered with Python using a custom orchestrator, specialized agents with parallelism (ThreadPoolExecutor), and statistical market analytics (average, standard deviation vs. benchmark, outlier detection).	Jan 2026
Data Stewardship Application Data quality and governance web application that evaluates CSV datasets against custom user-defined rules and automatically generates data dictionaries. Built with React 19, TypeScript, and Vite, leveraging Gemini 2.5 Flash, PapaParse for CSV parsing, and react-i18next for multi-language support (ES/EN).	Jan 2026
Credit Card Customer Churn Prediction Classification model developed to predict credit card customer churn. Evaluated Random Forest, AdaBoost, and SVM models using SMOTE for class balancing, StandardScaler, and cross-validation, measuring success via F1-score and confusion matrices (scikit-learn, imblearn). Formulated as a PoC study trained on public datasets.	Jan 2024

Banking Fraud Detection — Solution Design Architectural design of a hybrid banking fraud detection system combining a rules engine with machine learning anomaly detection. Proposed anomaly models (Isolation Forest, Autoencoders, One-Class SVM) and classification targets, optimizing precision/recall/F1 scores across point, contextual, and collective anomalies. Documented under a comprehensive design specification (v1.6).	2024-2025
Mutual Funds — Segmentation and Propensity (ICBC) Data-driven initiative to increase mutual fund (FCI) penetration among individual clients by segmenting customers and predicting propensity to invest. Defined target variables, extracted and processed millions of records from Teradata databases, designed predictive features, and planned campaign re-training and A/B testing workflows.	2024-2025
Agronomic Trial Statistical Analysis (Soybean) Conducted complete statistical analysis of a soybean agricultural trial. Performed exploratory data analysis (EDA), parametric/non-parametric testing, and mixed-effects modeling (statsmodels: OLS, mixedlm, ANOVA, Tukey HSD) using synthetic data, generating comprehensive PDF analysis reports.	Jan 2024
Exploratory Soil Data Analysis Exploratory Data Analysis (EDA) on 11,866 soil records (2021–2026) focusing on Mineralized Nitrogen (Nan) vs. Organic Matter, Total Nitrogen, and pH. Performed Pearson/Spearman/PPS correlations, non-linear mixed-effects modeling, month/season fixed effects, and IQR outlier detection.	2021-2026
Country Expansion Program Demographic segmentation. Forecast the Future Demand for actual cities. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Logistics prediction of the level of customer satisfaction Health Score for NPS. Anheuser-Busch InBev. Pi consulting & Strategies.	Jan 2022
Genotype by Environmental Predictions GDM. Don Mario. Data Science. Pi consulting & Strategies.	Jan 2021
Digital Government Project of the Ministry of Economy and Energy Data Science, Natural Language Process. Pi consulting & Strategies.	Jan 2021
Text Classifications to improve the management system Pernod Ricard. Data Science. Pi consulting & Strategies.	Jan 2021
PNCYO 1127046 Evaluations networks unpublished of germplasm INTA. Statistical Analysis of Genotypes regarding environments. Publications of the results. INTA.	2013-2019
PNBIO 1131042 Applied Genomics to molecular optimization parametrical and non-parametrical statistical models. INTA.	2013-2019
PNBIO 1131043 Bioinformatics and genomics. Assistance and new models in the evaluation of big base data. INTA.	2013-2019
CORDO 1262206 Transfer and training in the application in new experimental designs. INTA.	2013-2019

Publications

Agentic AI in Practice: Bridging the Gap between Econometrics and LLMs <i>Carolina Del Pilar Díaz</i> (LinkedIn Tech Talks)	Jan 2025
Advanced Statistical Architectures for Complex Data Demetrio C. G. B., <i>Díaz C. P.</i> (XXX Reunión Científica del GAB)	Jan 2025
The bright side of alfalfa viral diseases: The response of coumestrol phytoestrogen content in alfalfa genotypes Pilar Silva, M., Basigalup, D. H., <i>Carolina Del Pilar Díaz</i> (Plant Pathology)	Jan 2020
Segregations Analysis of Polymorphic SSR in three populations of F2 of peanut for charcoal tolerance Ritta L., Moreno M.V., Mamaní E., <i>Díaz C.</i> , Etchart V. (Peanut Workday)	Jan 2020
Advanced Mixed Generalized linear models, for effects of two levels of residues of Suico extracts on ruminal fermentation in vitro <i>Díaz, C. P.</i> , Scorcione Turcato, M.C, Ortiz Chura A. (XXIII Reunión Científica del GAB)	Jan 2019
The influence of insecticide applications on stored soybeans on the level of residues in the crude oil Aguilar Roxana, Nassetta Mirtha, <i>Díaz Carolina</i> , Rojas Dante (7th Congress International Food Science and Technology)	Jan 2019
Nutritional chemical composition of fruits of different carob species from northwestern Córdoba Martinez MJ, Ledesma, M, Aguilar R, <i>Díaz C.</i> , Nassetta M (VII International Congress of Food Sciences and Technology)	Jan 2019
Assessing INTA's Arachis hypogaea Core Collection for reaction to peanut smut Mamaní EMC, Rodríguez AV, Cordes G, <i>Díaz CDP</i> , Moreno MV (Conferencia del Genoma del maní)	Jan 2018
Corn Networks: Analysis of genotype-environment. Application of GGE ShuklaModels <i>Díaz Carolina Del Pilar</i> , Ferreyra M.Laura (Horizonte magazine)	Jan 2018
Alfalfa (Medicago sativa L.) coumestrol phytoestrogen content in response to genotypes, cultivar and viral disease Silva, M., Chiacchiera, S., Mamaní, E., <i>Díaz, C.</i> (Second World Alfalfa Congress)	Jan 2018
Regional Evaluation of INTA Wheat Varieties Cordes D., Candela R., <i>Díaz C.</i> (Rural exhibition of Jesús María)	Jan 2018

Cluster and Yield Evaluation for Corn Varieties Chincuini D., Druetta A., <i>Díaz C.</i> , Vega C. (Rural exhibition of Jesús María)	Jan 2018
Phenotyping for Drought Tolerance in Sunflower inbred lines Grandon N., Martin E., Cicconi E., <i>Díaz C. P.</i> (32nd Meeting of Plant Physiology Congress)	Jan 2018
Effect of Chemical and Biological control of acid vine rot Cordes, D.D., Grion, A., <i>Díaz C.</i> (40th Argentine Congress of Horticulture)	Jan 2018
Effect of Lippia turbinata G extracts, at two levels of inclusion, on in vitro ruminal fermentation Scorcione Turcato, M.C., Ortiz Chura A., <i>Díaz, C. D. P.</i> (41° Congreso de la Asoc. Argentina de Producción Animal)	Jan 2018
Deltamethrin residues in crude soybean oil Aguilar, R., <i>Díaz, C.</i> , Strada, J. (Periurban Conferences)	Jan 2017
Analysis of Chlorpyrifos methyl residues in crude soybean oil from treated grains in storage Aguilar R., Nassetta M., <i>Díaz C.</i> (IV Ibero-American Congress of Environmental Health)	Jan 2017
Dissipation dynamics of organophosphate insecticides in crude oil of soybean Aguilar R., <i>Díaz C.</i> , Strada J., Nassetta M. (V International Congress of Food Science and Technology)	Jan 2016
Effects of the alfalfa stunt viral complex on different genotypes estimated with Mixed Linear Models <i>Díaz, C.</i> , Silva, M., Mamaní, E., Basigalup, D. (XXI GAB Scientific Meeting)	Jan 2016
Genetic characterization of cultivated peanut genetic resources using microsatellite markers Pozzi, F, Etchart V, <i>Díaz C.</i> , Moreno, M. V (Rev. FCA UNCUYO)	Jan 2014

Certificates and Courses

- Agentic AI on AWS. BeSA- Become a Solution Architect. Candidate. (2026)
- Databricks Certified Generative AI Engineer Associate (2026)
- Microsoft Certified: Azure AI Fundamentals. Candidate. (2026)
- AWS Academy Cloud Architecting – Candidate, WIC Training Program (2025)
- AWS Academy Graduate — Cloud Foundations (2026)
- AWS Academy Graduate — Machine Learning Foundations (2026)
- IBM SkillsBuild — Certificado de finalización (2026)
- Open English – Candidate (Online Course) (2025-Present)
- AWS Academy Graduate - Cloud Foundations (2024)
- Generative AI Large Language Models & ChatGPT Prompt Engineering for Developers. Deep Learning AI. (2023)
- MLOps Fundamentals CI/CD/CT pipelines with Azure (2022)
- Databricks and Apache Spark for Big Data (2022)
- Machine Learning Pipelines with Azure ML Studio & Azure Synapse SQL Pool (2021)
- Bayesian Statistics: Techniques and Models. University of California (2021)
- AWS Academy Graduate - Machine Learning (2020)
- Diploma in Data Science Machine Learning and applications, National University of Córdoba (2020)
- Machine Learning by Professor Andrew Ng, Stanford University / Coursera (2018)
- Brief Course on Machine Learning & Brief Course on Mixed Models (2016)
- Post-Graduate Course of Bayesian Statistics, National University of Córdoba (2016)